

Datasets for Navigating Sensitive Topics in Recommendation Systems

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A Related Work

A.1 Datasets of Sensitive or Harmful Content

There are several types of sensitive or harmful content, and several academic datasets for studying their prevalence online. Content which spreads misinformation or conspiracy theories has received much attention. Faddoul et al. (2020) study the presence of conspiracy videos on YouTube over time, and Liaw et al. (2023) provide the YouNICON dataset of conspiracy YouTube videos. Nielsen and McConville (2022) provide Mumin, a dataset of fact checked misinformation on Twitter. This type of harmful content has also recieved attention from the recommender systems community: Fernández et al. (2021) and Tommasel and Menczer (2022) link misinformation labels with recommendation data to study amplification.

Another type of harmful content is hate speech or toxic content. Almerkhi et al. (2020) present a dataset of Reddit comments and toxicity labels. Mollas et al. (2020) present a dataset of hate speech in Reddit and YouTube comments. Wulczyn et al. (2017) provide a dataset of personal attacks in Wikipedia comments. For all of these works, the labels were generated by crowd workers.

The aforementioned types of harmful content are generally viewed as universally negative, and thus datasets are created and used with the implicit goal of limiting or preventing its proliferation. These goals can be fraught, since the precise definition of concepts like “hate speech” and “misinformation” are often contested and may suffer from biases against marginalized groups (Vidgen and Derczynski, 2020).

In contrast, our focus is on user generated content warnings, which are created by the same communities who use them as a tool to curate media consumption. As a result, this form of sensitive content is less relevant to censorship and more relevant to ensuring user agency. Wiegmann et al. (2023) provide the Webis Trigger Warning Corpus, an extensive dataset and taxonomy of trigger warnings which we leverage. Despite the relevance of sensitive content labels to real world recommendation systems like Instagram (Instagram, 2021) and TikTok (Goodman, 2020), we are not aware of recommendation focused datasets that provide sensitive content labels. We hope this work will fill that gap.

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A.2 Harm in Recommendation Systems

There is extensive work characterizing various types of harmful impacts from recommender systems. Shelby et al. (2023) propose a taxonomy of sociotechnical harms based on an extensive literature review. Five major types of sociotechnical harms are categorized: representational, allocative, quality of service, interpersonal, and social systems harms.

Some harms studied act at the societal level. Ribeiro et al. (2020) conduct an audit study of YouTube to study radicalization pathways. Ledwich et al. (2022) study filter bubbles on YouTube by analyzing the recommendations of stylized bots with content preferences and watch history. Levin et al. (2021) present several works on the dynamics of political polarization; mathematical models of polarization are formulated and theoretically analyzed. Restrepo et al. (2021) observe how Facebook parenting communities are pushed closer to extreme communities, and then use a dynamical systems model to derive strategies for controlling this type of amplification. Whittaker et al. (2021) conduct an empirical analysis of YouTube, Reddit, and Gab’s recommendation systems when interacting with far-right content, showing that YouTube amplifies extreme content, while Reddit and Gab do not. Gormann and Armstrong (2022) conducts theoretical and simulation studies on the consequences of failed alignment with human values. Other harms studied act at the individual level. Lin et al. (2016) survey adults age 19 to 32 about social media use and depression, finding that increased social media use correlates strongly with an increased odds of depression. Smith et al. (2022) use case studies to summarize common causes of algorithmic harm and their negative consequences.

Work on multi-objective recommender systems has in part been developed as a response to these observed harms. Zheng and Wang (2022) and Jannach (2022) provide surveys of this field. In addition to the standard engagement maximization, these multi-objective works also optimize for diversity (Vargas and Castells, 2011), fairness (Xiao et al., 2017), multi-stakeholder utility (Sürer et al., 2018), polarization (Suna and Nasraouia, 2021), and harm proxies (Singh et al., 2020). This line of work requires supplemental data to measure the additional objective that is co-optimized with engagement maximization. The datasets we provide will enable multi-objective work that balances harm as measured by unwanted exposure to sensitive content.

A.3 Evaluating Fairness, Bias, Amplification

Unfairness and bias are other consequences that may result from recommendation systems. These topics have received much attention from the academic community; see surveys by Chen et al. (2023) and Ekstrand et al. (2022). Many measures of bias and (un)fairness capture discrepancies between expressed user preferences and recommendations. Over- or under-recommending certain content categories constitutes *mis-calibrated* recommendations (Steck, 2018). Several works identify bias relating to item popularity (Abdollahpouri et al., 2019; Ekstrand et al., 2018a), item genre (Lin et al., 2019), and creator or user demographics (Ekstrand et al., 2018b,a; Shakespeare et al., 2020) in domains including movies, music, books. Another line of work investigates bias via accessibility using counterfactual metrics. Rather than measuring mis-calibrated recommendations, it is measured by what *could be* recommended (Akpınar et al., 2022; Dean et al., 2020; Curmei et al., 2021; Guo et al., 2021).

B Additional Information

Table 1: The 36 warning labels used in the AO3 dataset and the number of works marked with the warning.

Warning	#Works
pornography	174391
violence	42084
other-mental-health	40560
death	36123
other-sexual	35784
sexual-abuse	35005
abuse	32111
other-medical	21428
blood-gore	19299
abusive-language	18708
suicide	13113
childbirth	12132
child-abuse	11449
mental-illness	11344
addiction	11179
incest	9864
homophobia	8392
self-harm	7628
kidnapping	7561
other-aggression	5717
collective-violence	5351
procedures	4784
dysmorphia	3525
other-pregnancy	1984
other-abuse	1613
sexism	1606
other-discrimination	1595
racism	1159
miscarriage	930
animal-abuse	744
transphobia	705
abortion	561
religious-discrimination	398
ableism	390
classism	326
body-shaming	67

Table 3: The 137 Does the Dog Die? trigger warnings used in the ML-DDD dataset along with the number of works marked "Clear Yes" for the warning and number of works marked "Clear No".

Warning	#Works
Is there blood/gore	6121
Is there sexual content	5406
Is there gun violence	4588
Does a parent die	3743
Does someone struggle to breathe	3372
Does someone abuse alcohol	3089
Is there a hospital scene	3074
Is someone kidnapped	3045
Does it have a sad ending	2980
Does someone use drugs	2935
Is there a dead animal	2822
Is someone tortured	2762
Does someone fart or spit	2756
Does a car crash	2747
Is someone stalked	2734
Is there shaving/cutting	2630
Does someone cheat	2617
Is someone sexually assaulted	2557
Does an animal die	2495
Are there flashing lights or images	2395
Does someone vomit	2373
Is there hate speech	2370
Are there anxiety attacks	2293
Is there shakey cam	2215
Is a child abused	2169
Does a kid die	2167
Are there jumpscars	2161

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Warning	#Works	Warning	#Works
Is there domestic violence	2145	Are there ghosts	867
Does someone break a bone	2076	Is there misophonia	834
Is someone restrained	2027	Is there obscene language/gestures	791
Is someone gaslighted	2009	Does someone suffer from PTSD	785
Are needles/syringes used	1961	Are any teeth damaged	782
Is there a claustrophobic scene	1925	Is someone possessed	721
Is there audio gore	1912	Is there a mental institution scene	712
Is there a shower scene	1888	Are there snakes	668
Is there addiction	1864	Are there n-words	630
Does someone die by suicide	1840	Does a plane crash	620
Does a car honk or tires screech	1734	Are there spiders	602
Does someone fall to their death	1701	Is someone homeless	598
Are there bugs	1700	Does someone becomes unconscious	576
Are animals abused	1599	Is there childbirth	574
Is there ableist language or behavior	1599	Does an LGBT person die	544
Is someone burned alive	1508	Is there genital trauma/mutilation	539
Is someone hit by a car	1506	Does someone have cancer	539
Is someone sexually objectified	1342	Are there incestuous relationships	539
Does the dog die	1298	Does someone say "I'll kill myself"	524
Is there amputation	1292	Is there body dysmorphia	516
Is there finger/toe mutilation	1243	Does someone attempt suicide	513
Are there homophobic slurs	1238	Does someone have a seizure	494
Does someone self harm	1229	Is there antisemitism	485
Are there fat jokes	1208	Does someone wet/soil themselves	467
Are there nude scenes	1114	Is there a large age gap	465
Does a head get squashed	1109	Is a child's toy destroyed	437
Is a mentally ill person violent	1107	Is there cannibalism	436
Is there eye mutilation	1085	Does a pet die	431
Does someone drown	966	Are there "Man in a dress" jokes	420
Does a baby cry	965	Does someone fall down stairs	410
Are there babies or unborn children	944	Does a cat die	392
Is there excessive gore	907	Is the fourth wall broken	385
Is there a hanging	884	Does someone sacrifice themselves	379
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Continued from previous column		Continued from previous column	
Warning	#Works	Warning	#Works
Are there clowns	379	Is a baby stillborn	50
Is someone buried alive	376	Are there 9/11 depictions	38
Is someone misgendered	364	Are there fat suits	20
Is the R word used	360	Is there bisexual cheating	17
Does someone asphyxiate	348	Is there D.I.D. misrepresentation	14
Does the black guy die first	348	Does a non-human character die	13
Is there copaganda	342	Does the abused become the abuser	6
Is a minor sexualized	332	Is there bestiality	4
Does a pregnant person die	293	Is there body dysphoria	4
Is there body horror	281	Is there aphobia	3
Does someone have a chronic illness	273	Concluded	
Is there a nuclear explosion	266		
Is someone crushed to death	263		
Does someone have an eating disorder	249		
Does someone miscarry	243		
Is a minority is misrepresented	212		
Is electro-therapy used	209		
Is a male character ridiculed for crying	207		
Is a child abandoned by a parent	191		
Is someone raped onscreen	190		
Are there abortions	182		
Does someone overdose	166		
Are there razors	147		
Is there autism specific abuse	124		
Is an infant abducted	112		
Is someone drugged	105		
Is there dog fighting	105		
Is Santa (et al) spoiled	87		
Are there mannequins	83		
Does a dragon die	74		
Does someone have a stroke	68		
Is there dementia/Alzheimer's	53		
Is there Achilles Tendon injury	52		
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C Details on Recommendation Algorithms

We consider two non-personalized recommendation algorithms, Random and TopPop, as baselines for each dataset (Kille and Lommatzsch, 2019). The Random algorithm generates recommendations at random, drawing works a user has not interacted with from a uniform distribution. The TopPop algorithm sorts works by popularity and recommends the top- k that a user has not seen. Popularity is measured by the average rating for a movie in ML-DDD and the number of interactions with a work in AO3.

For a more personalized approach, we also trained a Collaborative Filtering algorithm on each dataset. Collaborative filtering is a technique used in recommendation systems to predict the preferences of a user by amassing data on other user-item interactions. It assumes that similar users will prefer similar items and uses this principle in generating ratings or recommendations.

Singular Value Decomposition (SVD) is a well-established matrix factorization technique (Mnih and Salakhutdinov, 2007). It performs collaborative filtering by constructing matrices of user and item latent factors to generate predictions. We apply SVD to the ML/DDD dataset using the Python library Surprise (Hug, 2020).

Unlike the explicit 0.5-5 scale ratings of ML-DDD, the AO3 dataset contains implicit user interactions from publicly available accounts awarding Kudos to a work. This means there is no clear notion of dislike in the dataset as it is not differentiated from works that a user simply never came across. For this reason, we use the Implicit (Frederickson, 2023) Python library’s implementation of an Alternating Least Squares (ALS) (Hu et al., 2008) algorithm catered for implicit data to generate recommendations for the AO3 dataset. This algorithm also uses matrix factorization for collaborative filtering, but deals with missing values by iteratively learning confidence levels to determine if the missing values indicate a negative or positive preference.

For ML-DDD, we utilize the entirety of the dataset (22.8M interactions) for conducting accuracy evaluations and generating recommendations for 1,000 random users to analyze amplification. For the evaluation of the AO3 dataset (i.e. calculating accuracy metrics), we use a representative subset of the data with 10% of the works and their interactions (3.9M interactions) due to computational constraints. However, we do use the complete dataset to train a model to generate recommendations for 1,000 random users and perform the amplification analysis.

For SVD, the algorithm’s factors, epochs, learning rate, and regularization were tuned to optimize rating prediction Root Mean Squared Error (RMSE) on a 90-10 train-test split. For Implicit matrix factorization, the algorithm’s factors, regularization, and weight of positive samples are tuned to optimize recall at $k=50$ recommendations. Hyper-parameters for both datasets can be found in the included code.

C.1 Accuracy Evaluation Metrics

To evaluate the performance of the algorithms, we draw from the standard predictive and classification accuracy metrics (Herlocker et al., 2004). For predictive accuracy, we report the Root Mean Squared Error (RMSE) of SVD, the only algorithm which produces rating predictions. RMSE measures the square root of the average squared difference between

the 0.5-5.0 scale ratings predicted by the algorithm and the true ratings for user-movie interactions. We split the rating data into a 90-10 train-test split and report the RMSE of the predictions on the test set. The split is made with respect to each user: 90% of the user’s interactions are in the trainset, and 10% in the testset.

However, the interaction data in AO3 is implicit: a 1 indicates a user liked an item, and a 0 indicates a user either does not like or has not seen an item. Rather than predicting this binary interaction, the algorithms trained on the AO3 data give recommendations directly. The TopPop and Random algorithms for ML-DDD give recommendations directly as well. To evaluate these algorithms, we report the classification accuracy metrics Precision, Recall, and F1 Metric at k recommendations for three values of k . We adhere by the definitions of these metrics in (Herlocker et al., 2004), summarized here:

- *Precision@k*. Fraction of relevant items in the k recommendations divided by k .
- *Recall@k*. Fraction of relevant items in the k recommendations divided by the total number of relevant items.
- *F1@k*. Two times the product of *Precision@k* and *Recall@k* divided by the sum of *Precision@k* and *Recall@k*.

To define relevancy and calculate the classification metrics, we use a methodology similar to Sarwar et al. (2000); Wilson et al. (2014). We split the rating data into a 90-10 train-test split with respect to each user: again, 90% of the user’s interactions are in the trainset, and 10% in the testset. After training each algorithm on the trainset, we generate k recommendations for each user that are unseen in their trainset interactions. The recommendations are then evaluated for precision and recall against the user’s relevant testset interactions. A relevant work in AO3 is one that appears in a user’s history, as all their interactions are positive. For ML/DDD, we consider movies in user’s history that are rated in their top-quartile of interactions as relevant, following suit with Basu et al. (1998).

C.2 Accuracy Results

Predictive Accuracy. The SVD algorithm has a RMSE of 0.7625 when trained then tested on the 90-10 random split of the data as described in Section 4.2. To generate novel recommendations for the 1,000 randomly selected users in the ML-DDD dataset, we train SVD on the entire dataset and evaluate its performance on the interactions of the random users. This produced a RMSE of 0.6733.

Classification Accuracy. The Precision, Recall, and F1 metrics at $k = 10, 50$, and 100 recommendations are displayed in Table 4. The Random algorithms perform significantly worse than both personalized recommenders on all metrics, which is expected of a simple baseline which does not consider any aspects of the data. In ML-DDD, the Random algorithm slightly outperforms TopPop, which may be due to the high interaction density of the dataset. Conversely, the TopPop algorithm performs quite well in AO3, outdoing both the Random and ALS algorithms. This indicates the users of AO3 are more likely to have interacted with popular content. SVD achieves higher accuracy than ALS, potentially due to the more telling explicit interactions of ML-DDD.

Table 4: Classification Accuracy Results for each algorithm.

	Precision @k=10	Precision @k=50	Precision @k=100	Recall @k=10	Recall @k=50	Recall @k=100	F1 @k=10	F1 @k=50	F1 @k=100
ML-DDD									
Random	1.00e-4	1.20e-4	4.00e-5	5.00e-4	2.67e-3	2.17e-3	1.67e-4	2.30e-4	7.85e-5
TopPop	1.00e-4	2.00e-5	2.00e-5	5.00e-4	5.00e-4	7.50e-4	1.67e-4	9.10e-5	9.40e-5
SVD	2.10e-3	4.40e-4	2.20e-4	8.48e-3	8.82e-3	8.82e-3	3.37e-3	8.38e-4	4.29e-4
AO3									
Random	4.64e-5	4.96e-5	4.86e-5	3.17e-4	1.71e-3	3.29e-3	8.09e-5	9.64e-5	9.59e-5
TopPop	4.82e-4	7.26e-4	6.69e-4	1.43e-3	5.99e-3	1.07e-2	7.22e-4	1.29e-3	1.26e-3
ALS	4.27e-4	1.00e-4	5.14e-5	2.13e-3	2.51e-3	2.57e-3	7.11e-4	1.93e-4	1.01e-4

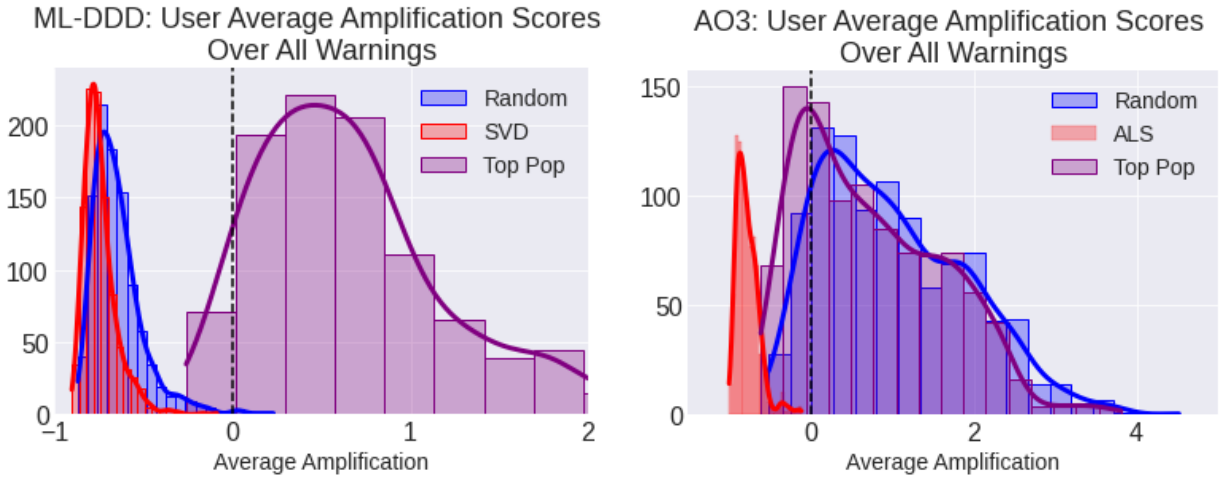


Figure 1: Distribution of amplification scores averaged over all warnings in ML-DDD (Top) and AO3 (Bottom) for each of the 1,000 randomly selected users. Amplification: the ratio of recommended works vs their history which have a given warning. The black dashed line is no amplification. In ML-DDD, TopPop produces the most amplification, and SVD the least. In AO3, Random produces the most amplification, and ALS the least.

D Initial Amplification Analysis

When considering the relationship between recommendation systems and content warnings, we introduce a novel metric called *Warning Amplification* inspired by metrics for algorithmic amplification (Huszár et al., 2021; Bouchaud, 2024). *Amplification@k* is defined as: for a given user, the fraction of items in their k top recommendations which have a given warning divided by the fraction of items in their history with the warning. We calculate this metric for each user in the randomly selected subset. We normalize this metric by subtracting 1, so 0% amplification indicates the user receives the same amount of warnings in their recommendations as exists in their history. To avoid dividing by zero in the case when the user does not have any items with the warning in their history, we hallucinate a single item with the given warning in the user’s history.

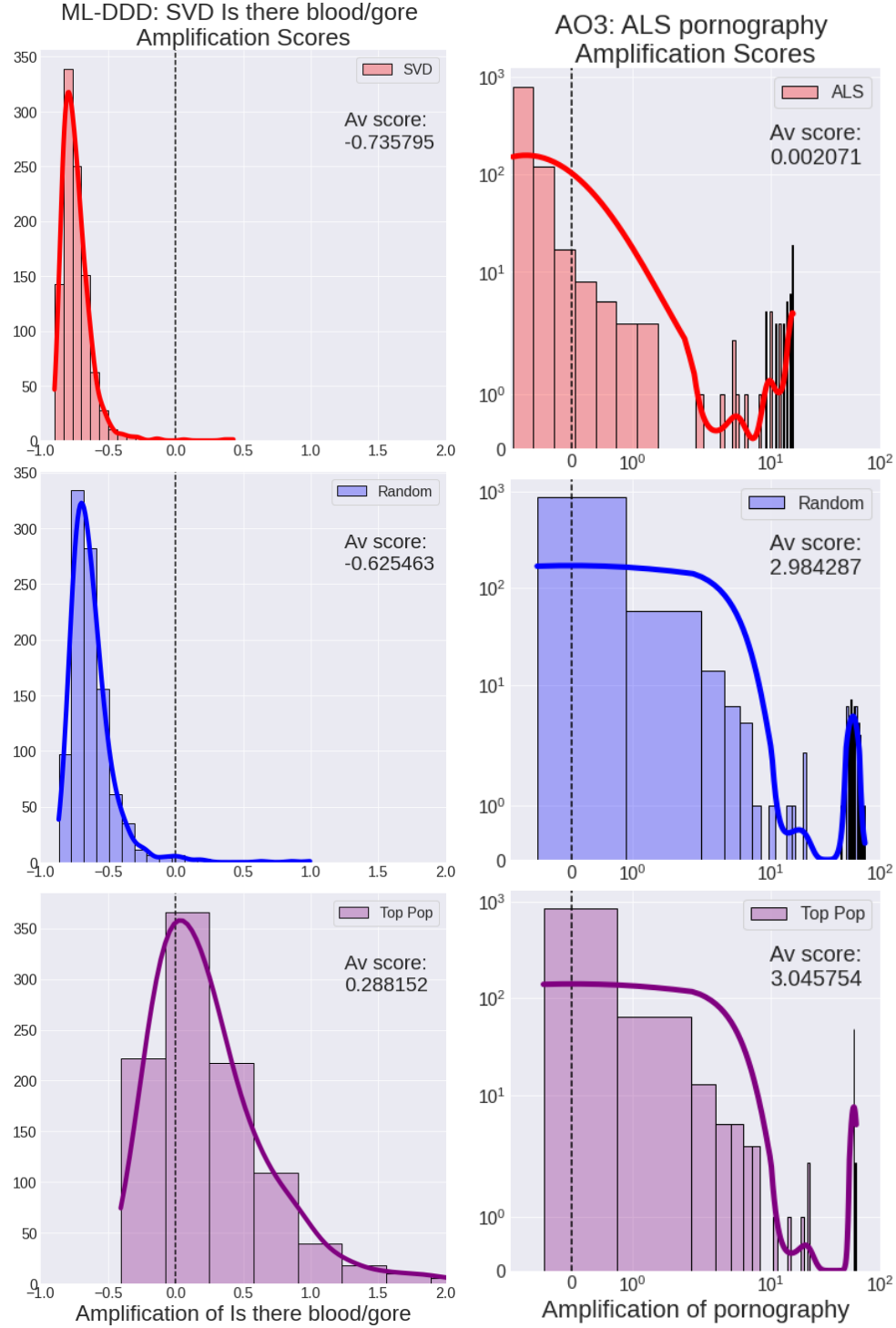


Figure 2: Amplification of the ML-DDD "Is there blood/gore" warning by the Random, TopPop, and SVD algorithms for $k = 100$ recommendations. The black dashed line represents 0% amplification, indicating the percent of works with blood/gore is the same in a user's history and their recommendations.

We present the distributions of user average amplification scores for $k = 100$ recommendations generated by each of the algorithms in Figure 1 (ML-DDD) and Figure 1 (AO3). A user’s average amplification is calculated by computing the mean of their Amplification@k scores for each warning. The black dashed line indicates where amplification is 0%, i.e. there is no difference in the amount of warnings in a user’s relevant history and their recommendations. In ML-DDD, we define relevant works as those in a user’s top quartile of ratings. In AO3, all works in a user’s history are relevant since they correspond to a positive interaction (kudos).

In Figure 1 we see that TopPop produces the most amplification and SVD the least. TopPop recommends popular works to users, and as examined in the permutation tests, a high proportion of warnings correlate positively with work popularity. Given the non-personalized nature of this algorithm, users are given popular yet warning-dense recommendations. The Random recommender chooses works at random, and given the low warning density of the dataset as noted in Section 3.1, it is not surprising that the algorithm produces less recommendations with warnings than those in a user’s relevant history. The SVD recommender has the lowest average amplification scores and recommends less items with a warning than in a user’s relevant history.

In Figure 1 we see the user average amplifications for AO3. The Random algorithm produces the most amplification and ALS the least. The Random algorithm likely amplifies warnings more than in ML-DDD given the higher warning density of the AO3 dataset. Similarly to the ML-DDD analysis, the personalized recommender amplifies warnings the least, consistently deamplifying sensitive content.

We also consider the average amplification of each warning, calculated by computing the mean of all user’s Amplification@k scores for the single warning. In Figure 2 we see a comparison of the amplification of "Is there blood/gore" by the ML-DDD algorithms and "pornography" by the AO3 algorithms. Following suit with the user average amplification analysis for ML-DDD, TopPop produces the most amplification of the blood/gore warning and SVD the least. The user average amplification trends for AO3 are also closely replicated on the pornography warning, but with TopPop amplifying slightly more than Random. From the permutation tests, we know blood/gore correlates negatively with popularity and pornography correlates positively. This is reflected in the figures as amplification for blood/gore for all algorithms is much lower than that of pornography.

Overall, our preliminary analysis of the relationship between sensitive categories and recommendation systems reveals how the personalized algorithms do not amplify warnings in comparison to baselines. This finding presents an interesting avenue for future research to explore, aiming to understand the nuances in the algorithms that lead to deamplification.

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